

AI POWERED CUSTOMER INTELLIGENCE TRANSFORMATION PROJECT

PART 4: NETWORK DIAGRAM & CRITICAL PATH ANALYSIS

Client: DOVV Distribution SA

1. Method

This analysis follows the standard forward pass / backward pass method for Activity-on-Arrow network scheduling, applied to the 42 tasks and their dependencies defined in the Project Schedule.

The forward pass computes Earliest Start (ES) and Earliest Finish (EF) for each activity by working from the project's first activities toward its last, taking the latest Finish Date among all of an activity's predecessors as the basis for its own Start Date. This is the same calculation used to build the Project Schedule and Gantt Chart.

The backward pass computes Latest Start (LS) and Latest Finish (LF) by working in the opposite direction, from the project's final activity back to its first, taking the earliest Latest Start among all of an activity's successors as the basis for its own Latest Finish. Total Float is then calculated as the difference between Latest Start and Earliest Start for each activity.

The full Network Diagram, presented as a separate document, shows all 42 activities with their ES, EF, LS, LF and Total Float values, connected according to the dependencies defined in the WBS and Project Schedule.

2. Activity Sequence and Dependencies

The network is not a single linear sequence. Three distinct chains run from the project's early activities to its final phase:

- Technical and adoption chain: Phase 2 (technical development) feeds into Phase 3 (adoption strategy) and Phase 5 (adoption campaign execution), converging into Phase 7 (pilot testing).
- Data and model chain: Phase 2 also feeds into Phase 4 (data collection) and Phase 6 (model training), which converges separately into Phase 7.
- Phase 7 itself requires both chains to be complete before its final activities (7.1 onward) can begin, since full system testing depends on both the adoption campaign's completion (Task 5.4c) and the trained model's integration (Task 6.3).

3. Critical Path

Result: The critical path runs through 20 of the 42 activities, all carrying zero Total Float: 2.2a → 2.2a-bis → 2.2b → 2.3a → 2.3b → 2.4 → 5.1 → 5.2 → 5.3 → 5.wait1 → 5.4a → 5.wait2 → 5.4b → 5.wait3 → 5.4c → 7.1 → 7.2b → 7.2c → 7.3 → 7.4.

This confirms, with a formal activity-by-activity calculation, the finding reached qualitatively during Week 2 scheduling: the adoption campaign chain, not the technical or data science work, determines the project's overall duration. The technical development activities within Phase 2 (2.2a through 2.4) are also critical, since they are the direct precondition for the adoption campaign that follows; however, the data collection and model training chain (Phase 4 and Phase 6) is not critical.

Total project duration, measured along the critical path, is approximately 333 calendar days from the 1 September 2026 start date, with the final activity (7.4) completing on 30 July 2027. This matches the timeline established in the Project Charter and confirmed in the Project Schedule and Gantt Chart.

4. Total Float and Scheduling Flexibility

The 22 activities not on the critical path carry positive Total Float, meaning they can be delayed without affecting the project's completion date, up to the limit of their float:

Activities	Total Float	Implication
Phase 4 activities (4.1 through 4.2c, including accumulation periods)	87 working days	Data collection can run substantially behind schedule without threatening the project deadline, as long as the model training chain that depends on it also has room to absorb the delay.
Phase 6 activities (6.1 through 6.3)	87 working days	Model training and validation, including the iterative bias-correction cycles anticipated in Task 6.2a, can extend well beyond their planned duration without schedule impact.
Phase 3 activities (3.1a through 3.2c)	29 to 38 working days	Adoption strategy design and communication material production have a comfortable margin, since they only need to be ready before the adoption campaign (Phase 5) begins.
Tasks 2.0, 2.1a, 2.1b	31 to 33 working days	The legal consent mechanism and card identification design can absorb delay without threatening the schedule, though both remain hard prerequisites for Task 2.4 and Task 5.1 respectively.
Task 2.2c	7 working days	The smallest float in the network outside the critical path itself, reflecting how closely this task trails the critical Phase 2 chain.
Task 7.2a	1 working day	A narrow margin caused by Task 7.2b (customer feedback collection) taking one day longer; Task 7.2c cannot begin until both are complete.

5. Why the Critical Path Is Important

The critical path is important for four concrete reasons specific to how this project will be executed and managed.

It defines where schedule risk actually lives

Any delay on a critical path activity delays the entire project, day for day. This directly reprioritises risk management established in the Week 1 Risk Register: Risk 2 (low customer adoption) and Risk 8 (POS provider documentation access, identified this week) both sit on the critical path, through the adoption campaign chain and Task 2.2a-bis respectively. Risk 6 (model underperformance), despite being rated High impact, sits entirely within the 87-day float of the data and model chain and therefore poses no threat to the project's completion date unless delays there exceed that margin.

It tells the Project Manager where to concentrate monitoring effort

With finite time and attention, monitoring cannot be uniform across 42 activities. The critical path identifies precisely which 20 activities warrant close, frequent tracking, and which 22 can be reviewed with a lighter touch, consistent with the routine communication cadence defined in the Week 1 Communication Plan.

It distinguishes urgency from importance

Model training (Phase 6) is arguably the most technically sophisticated part of this project, yet it is not on the critical path. The adoption campaign, comparatively simple in technical terms, is what determines whether the project finishes on time. This distinction prevents the natural tendency to over-manage the most complex-seeming work while under-managing the work that actually controls the deadline.

It provides an early warning threshold for corrective action

Because Total Float is calculated precisely for every activity, the Project Manager has a quantified answer to the question “how much delay can we absorb here before it matters” for every task in the schedule, rather than a general impression. A two-day slip on Task 2.2c, with 7 days of float, is a note for the next routine update. The same two-day slip on Task 5.3, with zero float, triggers the exception communication procedure defined in the Communication Plan and requires an immediate response.

6. Reference

The complete Network Diagram, showing ES, EF, LS, LF and Total Float for all 42 activities with the critical path highlighted, is provided as a separate document (Dovv_Network_Diagram.pdf).